

The weight spectrum of the Reed–Muller code $\text{RM}(7, 14)$ up to four weights

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Abstract

The binary Reed–Muller code $\text{RM}(7, 14)$ has minimum distance $d = 128$. Leuenberger and Albrizzio (arXiv:2606.21425) showed that every even number from 312 to $2^{14} - 312$ is a weight of $\text{RM}(7, 14)$, except possibly 22 values. We show that 14 of them are weights, by explicit codewords. We also give explicit codewords, checked by two independent programs, for all even numbers from 320 to $2^{14} - 320$ except 322, 326, 330, 334 and their complements. Hence an open question of Carlet on the weight spectra of $\text{RM}(m - c, m)$ has, for $(c, m) = (7, 14)$, a positive answer if and only if these four numbers are weights. Together with the theorem of Kasami, Tokura and Azumi on weights below $2.5d$, which we cite but could not consult, this determines the weight spectrum of $\text{RM}(7, 14)$ except for these four numbers and their complements; the equivalence with Carlet’s question does not depend on that theorem. Conjecture 2 of Lou and Wang, which would imply a positive answer, is false for $m = 6$. For the open weights we prove non-existence in restricted families and reduce each of them to a question about $\text{RM}(6, 13)$ and $\text{RM}(7, 13)$; calibrated searches found no codeword of weight 322, 326, 330 or 334, which is heuristic evidence only. The 17 explicit weights are formally verified in Lean 4. The results were found by AI coding agents working under the author’s direction; their role is described at the end of the paper.

1 Introduction

For $0 \leq r \leq m$, the binary Reed–Muller code $\text{RM}(r, m)$ consists of the truth tables on $\mathbb{F}_2^m$ of the polynomials in $x_1, \dots, x_m$ over $\mathbb{F}_2$ of degree at most r , where $x_i^2 = x_i$. We identify a codeword with its polynomial (its algebraic normal form), write $\text{wt } f$ for the number of points at which $f = 1$, and call the set of weights of the codewords the *weight spectrum*. The minimum distance of $\text{RM}(r, m)$ is $d = 2^{m-r}$. Kasami and Tokura [6] determined the weights below $2d$, and Kasami, Tokura and Azumi [7] those below $2.5d$. The whole weight spectrum of $\text{RM}(m - c, m)$ is known, for all $m \geq 2c$, for $c \leq 2$ (classical), for $c = 3, 4$ (Carlet and Solé [4]), for $c = 5$ (Carlet [2]) and for $c = 6$ (Lou and Wang [9]). Carlet asked [2] whether, for every $c \geq 1$ and $m \geq 2c$, the weight spectrum of $\text{RM}(m - c, m)$ consists of 0, 2^m , the weights in $[2^c, 2^{c+1}]$ given by Kasami and Tokura and in $[2^{c+1}, 2^{c+1} + 2^{c-1}]$ given by Kasami, Tokura and Azumi, their complements to 2^m , and *all* even numbers in $[2^{c+1} + 2^{c-1}, 2^m - 2^{c+1} - 2^{c-1}]$. (We paraphrase the wording of Carlet’s extended abstract [1] of [2], which is also that of [8, Conjecture 1]; we could not access the full text of [2]. In the earlier conjecture of Carlet and Solé [4, Section 6] the last interval starts at $3 \cdot 2^c$.) Since $d = 2^c$, the last interval starts at $2.5d$. The first open case is $(c, m) = (7, 14)$: here $d = 128$, and the question asks whether every even number from 320 to $2^{14} - 320 = 16064$ is a weight of $\text{RM}(7, 14)$.

Leuenberger and Albrizio [8, Proposition 4] showed that every even number from 312 to $2^{14} - 312$ is a weight of $\text{RM}(7, 14)$, except possibly the 22 elements of

$$M = \{322, 326, 330, 334, 354, 8174, 8178, 8182, 8186, 8188, 8190, \\ 8194, 8196, 8198, 8202, 8206, 8210, 16030, 16050, 16054, 16058, 16062\};$$

parts of their proof rest on computer searches whose details are not given (Lemmas 6 and 9 of [8]). They were unsure whether this is the whole spectrum, and remarked that their constructions would produce all of M from the eight weights 322, 326, 330, 334, 354, 4378, 4380, 4382. Their constructions follow Lou and Wang, whose Conjecture 2 [9] would give 322, 326, 330, 334 and 354 (see Theorem C).

Results. Throughout, we write

$$U_0 = \{322, 326, 330, 334\}, \quad U = U_0 \cup (2^{14} - U_0) = U_0 \cup \{16050, 16054, 16058, 16062\}, \\ K = \{0, 128, 192, 224, 240, 248, 252, 254, 256, 272, 288, 296, 304, 308, 312, 314, 316, 318\};$$

by the theorem of Kasami, Tokura and Azumi, K is the set of weights of $\text{RM}(7, 14)$ below 320; see (P4a) in Section 2.

Theorem A. *The following 17 numbers are weights of $\text{RM}(7, 14)$: 354, 4378, 4380, 4382, 8174, 8178, 8182, 8186, 8188, 8190, 8194, 8196, 8198, 8202, 8206, 8210 and 16030. Table 1 gives codewords with at most seven monomials.*

Fourteen of these numbers, the elements of $M \setminus U$, are listed as missing in [8]; to our knowledge they are new. The other three, 4378, 4380 and 4382, are not new: they are weights by [8, Proposition 4], and the remark of [8] lists them among the eight weights from which its constructions would produce M . We include codewords for them because we do not use [8, Proposition 4].

Theorem B. *Let $S_0 = (K \cup \{320, 322, \dots, 16064\} \cup (2^{14} - K)) \setminus U$. The weight spectrum of $\text{RM}(7, 14)$ is $S_0 \cup X$ for a set $X \subseteq U$ with $X = 2^{14} - X$. Every element of S_0 is the weight of an explicit codeword with at most eight monomials.*

In Carlet's question the weights below $2.5d$ and their complements are, by definition, those given by [6, 7], and all weights are even by (P1). So for $(c, m) = (7, 14)$ the question asks exactly whether every even number from 320 to 16064 is a weight. By the explicit codewords of Theorem B and by (P1), it has a positive answer if and only if 322, 326, 330 and 334 are weights. This equivalence does not depend on the theorem of Kasami, Tokura and Azumi; only the determination of the whole spectrum in Theorem B does, where that theorem is the only external input: it excludes weights below 320 outside K (see (P4a)). We could not consult [7] ourselves.

Construction 2 of [9] takes $g_1 = x_1 \cdots x_m + x_{i_1} \cdots x_{i_m} + x_{i_{m+1}} \cdots x_{i_{2m}}$ and $g_2 = x_1 \cdots x_m + x_{i_{2m+1}} \cdots x_{i_{3m}} + x_{i_{3m+1}} \cdots x_{i_{4m}}$ with $1 \leq i_j \leq 2m$, and $a_1, a_2 \in \mathbb{F}_2$, and forms the concatenation $g = 0 \parallel (g_1 + a_1) \parallel g_2 \parallel (g_1 + g_2 + a_2) \in \text{RM}(m+1, 2m+2)$ of four truth tables (Section 4). Conjecture 2 of [9] states that, for $m \geq 4$, the weights of these codewords include $2^{m+2} + 2^m + 2i$ and $2^{2m} + 2^m + 2i$ for $0 \leq i < 2^m$; for $m = 6$ the first family is $\{320, 322, \dots, 446\}$.

Theorem C. *Conjecture 2 of [9] is false for $m = 6$: no codeword of Construction 2 with $m = 6$ has weight 322. This remains true if g_1 and g_2 range over all of $\text{RM}(6, 12)$, as in the proof of [9, Proposition 5].*

These codewords even miss all of U (Proposition C'), and so do the sums of at most three indicators of flats of dimension at least 7 (Proposition D) and the sums of at most five monomials of degree at most 7 (Proposition E, by exhaustive computation). Proposition F reduces each $w \in U_0$ to a question about $\text{RM}(6, 13)$ and $\text{RM}(7, 13)$. Searches based on it found no codeword of weight in U_0 , although on $\text{RM}(6, 12)$ they find the analogues 162, 166, 170, 174 of U_0 within two minutes; this is heuristic evidence only (Section 6.2). Theorem A is formally verified in Lean 4 (Section 7).

Prior work. We found no earlier claim that an element of $M \setminus U$ is a weight of $\text{RM}(7, 14)$, no determination of its spectrum, and no disproof of Conjecture 2 of [9]. We searched on 26 September 2026, 06:41–07:15 and 13:43–14:03 UTC: the arXiv (listing of [8], still at version 1 at 13:43 UTC, and API searches); the works citing [8, 9, 2, 4] in Semantic Scholar; Crossref; zbMATH Open; the MathSciNet reference lookup; MathDB, whose entries on Carlet's question and on Conjecture 2 of [9] were open, without solutions or comments; MathOverflow and Mathematics Stack Exchange; the IACR ePrint archive; HAL; GitHub, including `trureturing` [14] and other repositories of AI-assisted work on open problems; and web search. Citation queries to OpenAlex and keyword searches of Semantic Scholar hit rate limits; the full texts of [7], [2] and the published version of [9] were not accessible, so we quote [9] and [8] from their arXiv versions 1; we did not search Google Scholar, and Scopus was not accessible. One possibly relevant paper could not be read: the note [10] of Lou and Wang (online 4 April 2026), whose title refers to two conjectures about the weight spectra of Reed–Muller codes; no abstract was available, its reference list in Crossref includes [2, 4, 6, 7, 9], and [8] does not cite it. It may concern the conjectures of [9], and so may anticipate Theorem C or other results here. A final check on 26 September 2026 between 15:49 and 15:52 UTC (the arXiv, where [8] still had only version 1; GitHub; MathDB; and MathOverflow) found nothing new. This reports the coverage of our searches, not a guarantee of priority.

2 Preliminaries

A k -flat is an affine subspace of dimension k , with 2^k points; its indicator 1_F is a product of $m - k$ affine functions, so it lies in $\text{RM}(m - k, m)$. A nonempty intersection of flats is a flat. We write $x_S = \prod_{i \in S} x_i$ ($x_\emptyset = 1$). Affine bijections preserve degrees, weights and flats. A *quadratic function* has degree at most 2. We use the following known results.

- (P1) If $r < m$, every codeword of $\text{RM}(r, m)$ has even weight, since $\text{wt } f \bmod 2$ is the coefficient of $x_1 \cdots x_m$ in f . As $\text{wt}(f + 1) = 2^m - \text{wt } f$, the weight spectrum is invariant under $w \mapsto 2^m - w$.
- (P2) The weights of $\text{RM}(r, m)$ are divisible by $2^{\lfloor (m-1)/r \rfloor}$ (McEliece [13]). In particular the weights of $\text{RM}(6, 13)$ are divisible by 4.
- (P3) The codewords of weight d are the indicators of $(m - r)$ -flats [11, Chapter 13]. By the theorem of Kasami and Tokura [6], in the form stated in [3, Section 2], every codeword of $\text{RM}(r, m)$, $r \geq 2$, with weight in $[d, 2d]$ is affinely equivalent to $x_1 \cdots x_{r-2}(x_{r-1}x_r + x_{r+1}x_{r+2} + \cdots + x_{r+2l-3}x_{r+2l-2})$ with $2 \leq 2l \leq m - r + 2$, or to $x_1 \cdots x_{r-l}(x_{r-l+1} \cdots x_r + x_{r+1} \cdots x_{r+l})$ with $3 \leq l \leq \min(r, m - r)$; both have weight $2d - 2^{m-r+1-l}$. Hence in $\text{RM}(6, 12)$ ($d = 64$) a codeword of weight 96 is $1_G q$ with G an 8-flat and q quadratic; one of weight 112 is of this form or is $1_F + 1_{F'}$ with 6-flats F, F' , $|F \cap F'| = 8$; and one of weight 126 is $1_F + 1_{F'}$ with 6-flats F, F' , $|F \cap F'| = 1$.

w	codeword f_w	$(\sigma_1, \dots, \sigma_k)$
354	$\mathbf{a} + \mathbf{b} + x_1x_2x_3x_5x_6x_7x_{11} + x_1x_3x_6x_7x_{10}x_{12}x_{13} + x_3x_4x_6x_7x_{10}x_{12}x_{13}$	(640, 203, 50, 12, 1)
4378	$\mathbf{a} + \mathbf{b} + x_3x_4x_6 + x_1x_5x_7x_{11}x_{13} + x_6x_{14}$	(6912, 1537, 173, 21, 1)
4380	$x_2x_6x_7x_9x_{12} + x_3x_4x_5x_8x_9x_{10}x_{13} + x_8x_{14} + x_1x_3x_5x_9x_{10}x_{11}x_{14}$	(4864, 288, 25, 1)
4382	$\mathbf{a} + \mathbf{b} + x_1x_2x_5x_6x_7 + x_4x_9 + x_3x_4x_6x_8x_{12}$	(5376, 725, 150, 20, 1)
8174	$\mathbf{a} + \mathbf{b} + x_7 + x_5x_{10} + x_2x_3x_4x_5x_8x_{10}x_{11} + x_1x_2x_4x_6x_{13}x_{14}$	(12928, 2825, 314, 59, 9, 1)
8178	$\mathbf{a} + \mathbf{b} + x_7 + x_2x_5x_6x_{13} + x_2x_5x_8x_{10}x_{13}x_{14}$	(9728, 1081, 187, 19, 1)
8182	$\mathbf{a} + \mathbf{b} + x_8 + x_1x_2x_3x_4x_6x_{11}x_{13} + x_1x_{14} + x_2x_7x_8x_{10}x_{11}x_{12}x_{14}$	(12800, 2765, 304, 50, 8, 1)
8186	$\mathbf{a} + \mathbf{b} + x_{11} + x_4x_{12} + x_1x_2x_8x_{10}x_{11}x_{12}x_{13}$	(12672, 2597, 219, 23, 1)
8188	$x_1 + x_2x_5x_6x_8x_9x_{10}x_{13} + x_2x_{14} + x_1x_4x_6x_7x_{11}x_{12}x_{14}$	(12544, 2370, 100, 2)
8190	$\mathbf{a} + \mathbf{b} + x_2 + x_1x_7x_8$	(10496, 1313, 82, 1)

Table 1: The codewords of Theorem A; $\mathbf{a} = x_1 \cdots x_7$, $\mathbf{b} = x_8 \cdots x_{14}$. If $f_w = \sum_{i=1}^k x_{S_i}$, then $\sigma_j = \sum_{|T|=j} 2^{14-|\bigcup_{i \in T} S_i|}$ over $T \subseteq \{1, \dots, k\}$, and $\text{wt } f_w = \sum_j (-2)^{j-1} \sigma_j$ (Lemma 2.1). The complements $1 + f_w$ for $w = 354, 8190, 8188, 8186, 8182, 8178, 8174$ have the weights 16030, 8194, 8196, 8198, 8202, 8206, 8210.

(P4a) The weights of $\text{RM}(7, 14)$ below 320 are exactly the elements of K , by the theorem of Kasami, Tokura and Azumi [7, pp. 392 ff.].¹

(P4b) The weights of $\text{RM}(6, 12)$ below 160 are 0, 64, 96, 112, 120, 124, 126, 128, 136, 144, 148, 152, 154, 156 and 158 ([9, Lemma 2], which cites [6, 7]).

Lemma 2.1. For $S_1, \dots, S_k \subseteq \{1, \dots, m\}$,

$$\text{wt} \left(\sum_{i=1}^k x_{S_i} \right) = \sum_{\emptyset \neq T \subseteq \{1, \dots, k\}} (-2)^{|T|-1} 2^{m-|\bigcup_{i \in T} S_i|}.$$

In particular, the weight depends only on the sizes of the $2^k - 1$ Venn regions of $S_1, \dots, S_k$.

Proof. For $y_1, \dots, y_k \in \{0, 1\}$ the exclusive or of the y_i equals $\sum_{\emptyset \neq T} (-2)^{|T|-1} \prod_{i \in T} y_i$, by induction on k from $y \oplus y' = y + y' - 2yy'$. Put $y_i = x_{S_i}(x)$ and sum over $x \in \mathbb{F}_2^m$: the product over T equals 1 exactly when $x_j = 1$ for all $j \in \bigcup_{i \in T} S_i$, which happens at $2^{m-|\bigcup_{i \in T} S_i|}$ points. The size of a union is a sum of sizes of Venn regions. $\square$

Lemma 2.2. Let $f: \mathbb{F}_2^{m+1} \rightarrow \mathbb{F}_2$, and put $g(x) = f(x, 0)$ and $g'(x) = f(x, 1)$ for $x \in \mathbb{F}_2^m$. Then $f = g + x_{m+1}(g + g')$, and $f \in \text{RM}(r+1, m+1)$ if and only if $g \in \text{RM}(r+1, m)$ and $g + g' \in \text{RM}(r, m)$. Moreover $\text{wt } f = \text{wt } g + \text{wt } g' = \text{wt}(g + g') + 2|\text{supp } g \cap \text{supp } g'|$, and $\text{supp } g \cap \text{supp } g' = \text{supp } g \setminus \text{supp}(g + g')$.

Proof. The identity holds at $x_{m+1} = 0$ and at $x_{m+1} = 1$. The polynomials g and $x_{m+1}(g + g')$ have no monomial in common, so $\deg f \leq r+1$ if and only if $\deg g \leq r+1$ and $\deg(g + g') \leq r$. The rest is counting. $\square$

3 Explicit codewords: Theorems A and B

Proof of Theorem A. Each f_w in Table 1 is a sum of distinct monomials of degree at most 7, so $f_w \in \text{RM}(7, 14)$. By Lemma 2.1, $\text{wt } f_w = \sum_j (-2)^{j-1} \sigma_j$ with the σ_j of the table; for instance

¹We could not consult [7] ourselves; the page reference is the one given by Carlet [1, Section 1] and by Carlet and Solé [4, Section 6]. No source we read gives the complete list of the weights of $\text{RM}(7, 14)$ between 256 and 320; Carlet [3, Section 2] says that the weights between $2d$ and $2.5d$ are too numerous to be recalled. Leuenberger and Albrizzio [8, Proposition 4] assert only that the elements of K are weights, not that there are no others below 320. The part of (P4a) below $2d = 256$ follows from (P3), and the computation after Proposition E agrees with (P4a).

$\text{wt } f_{354} = 640 - 2 \cdot 203 + 4 \cdot 50 - 8 \cdot 12 + 16 = 354$. The other seven weights are those of the complements $1 + f_w$, by (P1). All 17 weights were also checked by machine (Sections 6 and 7). $\square$

Remark 3.1. The weights 8174 to 8210 also follow from a direct sum: if A depends only on $x_1, \dots, x_7$ and B only on $x_8, \dots, x_{14}$, then $\deg(A+B) \leq 7$ and $\text{wt}(A+B) = \text{wt } A(128 - \text{wt } B) + (128 - \text{wt } A) \text{wt } B = 8192 - 2(64 - \text{wt } A)(64 - \text{wt } B)$. As $\text{wt } A$ and $\text{wt } B$ can take any value from 0 to 128, taking $\text{wt } B = 63$ and $\text{wt } A = 64 - n$ shows that every even number $8192 - 2n$ with $|n| \leq 64$ is a weight.

Proof of Theorem B. The file `rm714_all_witnesses.json` of [5] lists, for each of the 7901 elements w of S_0 , a sum of at most eight distinct monomials of degree at most 7 with weight w , as two independent programs confirm (Section 6). So every element of S_0 is a weight. Conversely, by (P1) every weight is even; by (P4a) the weights below 320 lie in K , and by complementation the weights above 16064 lie in $2^{14} - K$. So the spectrum lies in $S_0 \cup U$, and it equals $S_0 \cup X$ with X the set of weights in U ; $X = 2^{14} - X$ by (P1). $\square$

Most entries of the file were found by random sampling of sparse polynomials, and 362 by [8, Lemma 8]; this plays no role in the proof. As S_0 consists of the weights asserted in [8, Proposition 4] and the new weights of Theorem A, the file also checks that proposition independently.

4 Construction 2 of Lou and Wang: Theorem C

Following [9], for functions f_1, f_2, f_3, f_4 on $\mathbb{F}_2^{12}$ let $f_1 \parallel f_2 \parallel f_3 \parallel f_4$ be the function on $\mathbb{F}_2^{14}$ equal to f_1, f_2, f_3, f_4 on the cosets $(x_{13}, x_{14}) = (0, 0), (1, 0), (0, 1), (1, 1)$, so that its weight is $\sum \text{wt } f_i$. For $A, B \in \text{RM}(6, 12)$ we have $0 \parallel A \parallel B \parallel (A + B) = x_{13}A + x_{14}B \in \text{RM}(7, 14)$.

Lemma 4.1. *Let $A, B \in \text{RM}(6, 12)$ and $f = x_{13}A + x_{14}B$. Then $\text{wt } f = \text{wt } A + \text{wt } B + \text{wt}(A+B)$. If $\text{wt } f \in U_0$, then $\text{wt } f \neq 322$, and the multiset $\{\text{wt } A, \text{wt } B, \text{wt}(A+B)\}$ is $\{64, 126, 136\}$ if $\text{wt } f = 326$, $\{64, 112, 154\}$ if $\text{wt } f = 330$, and one of $\{64, 112, 158\}$, $\{64, 126, 144\}$, $\{96, 112, 126\}$ if $\text{wt } f = 334$.*

Proof. The first claim is the weight of the concatenation. Each of $A, B, A+B$ is the sum of the other two, and $\text{supp}(Y+Z) \subseteq \text{supp } Y \cup \text{supp } Z$, so each of the three weights is at most the sum of the other two. Hence, writing $a \leq b \leq c$ for the three weights, $c \leq a+b$ and $c \leq \text{wt } f/2 \leq 167$. By (P1) and (P4b), a, b, c lie in

$$L = \{0, 64, 96, 112, 120, 124, 126, 128, 136, 144, 148, 152, 154, 156, 158, 160, 162, 164, 166\}.$$

The triples in L with $a \leq b \leq c \leq a+b$ and $a+b+c \in U_0$ are the ones listed (certificate `cert_arith`, Section 6). For $\text{wt } f = 322$ this is also easy to see by hand. Then $c \leq 161$, so $a, b, c \leq 160$. The elements of L that are 2 modulo 4 are 126, 154 and 158, and $322 \equiv 2 \pmod{4}$, so an odd number of a, b, c are among them; as three of them sum to at least 378, exactly one is. But none of $322 - 126 = 196$, $322 - 154 = 168$ and $322 - 158 = 164$ is a sum of two elements of $\{0, 64, 96, 112, 120, 124, 128, 136, 144, 148, 152, 156, 160\}$. $\square$

Proof of Theorem C. In both readings $g_1, g_2 \in \text{RM}(6, 12)$, so let $g_1, g_2 \in \text{RM}(6, 12)$ be arbitrary and $g = 0 \parallel (g_1 + a_1) \parallel g_2 \parallel (g_1 + g_2 + a_2)$. If $a_1 = a_2 = 0$, then $g = x_{13}g_1 + x_{14}g_2$, and $\text{wt } g \neq 322$ by Lemma 4.1. If $a_1 = a_2 = 1$, then $g = 0 \parallel G_1 \parallel g_2 \parallel (G_1 + g_2) = x_{13}G_1 + x_{14}g_2$ with $G_1 = g_1 + 1 \in \text{RM}(6, 12)$, and the same holds. If $a_1 = 1$ and $a_2 = 0$, then $\text{wt } g = (4096 - \text{wt } g_1) + \text{wt } g_2 + \text{wt}(g_1 + g_2) \geq 4096$, as $\text{wt } g_2 + \text{wt}(g_1 + g_2) \geq \text{wt } g_1$; similarly $\text{wt } g \geq 4096$ if $a_1 = 0$ and $a_2 = 1$. So 322 is not a weight of Construction 2 for $m = 6$, whereas Conjecture 2 of [9] asserts that $2^8 + 2^6 + 2 = 322$ is one. $\square$

Proposition C'. *Let $A, B \in \text{RM}(6, 12)$. Then $\text{wt}(x_{13}A + x_{14}B) \notin U$. Consequently no codeword of Construction 2 with $m = 6$ has weight in U , whether g_1 and g_2 are as in [9] or arbitrary in $\text{RM}(6, 12)$.*

Proof. We use the fact (*): if G, E are flats of $\mathbb{F}_2^{12}$ and q is quadratic, then $|G \cap E \cap \text{supp } q|$ is even or at most 3, since q restricted to the flat $G \cap E$ (if nonempty, of dimension k) has even weight if $k \geq 3$ by (P1), and weight at most $2^k \leq 4$ otherwise.

The weight of $f = x_{13}A + x_{14}B$ is at most $3 \cdot 2^{12} = 12288 < 16050$, and it is not 322 by Lemma 4.1. For the five remaining triples of Lemma 4.1 we choose two of the functions $A, B, A + B$, say X and Y , so that $X + Y$ is the third; then $|\text{supp } X \cap \text{supp } Y| = (\text{wt } X + \text{wt } Y - \text{wt}(X + Y))/2$. We use (P3) throughout.

The triples $\{64, 126, 136\}$ and $\{64, 126, 144\}$. Take $X = 1_E$ with E a 6-flat and $Y = 1_F + 1_{F'}$ with $F \cap F' = \{p\}$. Then $|E \cap \text{supp } Y| = 27$, resp. 23. As $\text{supp } Y = (F \cup F') \setminus \{p\}$, $|E \cap F| + |E \cap F'| = |E \cap \text{supp } Y| + 2[p \in E]$ is 27 or 29, resp. 23 or 25. Both summands are 0 or powers of 2, so one of them is 1, and the other is 22, 24, 26 or 28, which is impossible.

The triples $\{64, 112, 154\}$ and $\{64, 112, 158\}$. Take $X = 1_E$ with E a 6-flat and $\text{wt } Y = 112$. Then $|E \cap \text{supp } Y| = 11$, resp. 9. If $Y = 1_G q$, this contradicts (*). Otherwise $Y = 1_F + 1_{F'}$ with $|F \cap F'| = 8$. Put $s = |E \cap F|$, $s' = |E \cap F'|$ and $t = |E \cap F \cap F'|$, each 0 or a power of 2. Then $s + s' - 2t$ is 11, resp. 9. It is odd, so $s = 1$ or $s' = 1$; by symmetry let $s = 1$. Then $t \leq 1$, and $s' = 10 + 2t$, resp. $s' = 8 + 2t$, so we are in the second case with $s' = 8$ and $t = 0$. By (P3) we may take $F = \{x_1 = \dots = x_6 = 1\}$ and $F' = \{x_1 = x_2 = x_3 = x_7 = x_8 = x_9 = 1\}$ (the form $x_1 x_2 x_3 (x_4 x_5 x_6 + x_7 x_8 x_9)$). Since $|E \cap F| = 1$, the affine map $E \rightarrow \mathbb{F}_2^6$, $x \mapsto (x_1, \dots, x_6)$, has a fibre with one point, so it is injective and hence bijective. So the set E_3 of points of E with $x_1 = x_2 = x_3 = 1$ has 8 points. As $E \cap F' \subseteq E_3$ and $|E \cap F'| = 8$, we get $E \cap F' = E_3$. The point of $E \cap F$ lies in E_3 , hence in F' , so $t = 1$, a contradiction.

The triple $\{96, 112, 126\}$. Take $X = 1_G q$ of weight 96 and $Y = 1_F + 1_{F'}$ of weight 126 with $F \cap F' = \{p\}$, so that $\text{wt}(X + Y) = 112$. Then $|\text{supp } X \cap \text{supp } Y| = 55$, and $|\text{supp } X \cap F| + |\text{supp } X \cap F'|$ is 55 or 57. One summand is odd, hence at most 3 by (*). The other is at most 48. Indeed, it is the weight of q on the flat $G \cap F$ (or $G \cap F'$). If this flat has dimension at most 5, the weight is at most 32. Otherwise $F \subseteq G$, and q is not constant 1 on F : otherwise $\text{supp } X \supseteq F$, and $X + 1_F$ would be a nonzero codeword of $\text{RM}(6, 12)$ of weight $32 < 64$. So $1 + q$ restricted to F is a nonzero codeword of $\text{RM}(2, 6)$, of weight at least 16, and q has weight at most 48 on F . The total is at most $51 < 55$, a contradiction.

Finally, in Construction 2 the cases $a_1 = a_2$ give codewords $x_{13}A + x_{14}B$ with $A, B \in \text{RM}(6, 12)$, as in the proof of Theorem C, and the other cases give weights between 4096 and 12288. $\square$

In particular, the codewords of [8, Constructions 1–3, Propositions 1 and 2], which are of this form, have no weight in U .

5 Partial non-existence results for the open weights

5.1 Sums of flats

Call two flats *transversal* if they meet in exactly one point. Two transversal flats of $\mathbb{F}_2^{14}$ of dimension at least 7 both have dimension 7: if their direction spaces are V and V' , then $V \cap V' = \{0\}$, so $\dim V + \dim V' \leq 14$.

Proposition D. *Let F_1, F_2, F_3 be flats of $\mathbb{F}_2^{14}$ of dimension at least 7, not necessarily distinct (the whole space is allowed). Then none of $\text{wt } 1_{F_1}$, $\text{wt}(1_{F_1} + 1_{F_2})$ and $\text{wt}(1_{F_1} + 1_{F_2} + 1_{F_3})$ lies in U .*

Proof. The weight of $1_{F_1} + 1_{F_2} + 1_{F_3}$ is $\sum |F_i| - 2 \sum_{i < j} |F_i \cap F_j| + 4|F_1 \cap F_2 \cap F_3|$, and a nonempty intersection of flats has a power of 2 as its size. As $4 \mid |F_i|$, the weight is 2 modulo 4 if and only if an odd number of the three pairs are transversal. The same holds for one or two flats. All elements of U are 2 modulo 4.

For one flat there is no transversal pair. Two transversal 7-flats give the weight $128 + 128 - 2 = 254$. Three pairwise transversal 7-flats give $384 - 6 + 4|F_1 \cap F_2 \cap F_3| \in \{378, 382\}$. It remains to treat exactly one transversal pair, say $F_1 \cap F_2 = \{p\}$ with $\dim F_1 = \dim F_2 = 7$, and $\dim F_3 = d \in [7, 14]$. Translate so that $p = 0$; then $F_1 = V_1$ and $F_2 = V_2$ are linear subspaces with $V_1 \oplus V_2 = \mathbb{F}_2^{14}$. Let V_3 be the direction space of F_3 . The pairs (F_1, F_3) and (F_2, F_3) are not transversal.

Case $p \in F_3$. Then $F_3 = V_3$, $|V_1 \cap V_3| = 2^a$ and $|V_2 \cap V_3| = 2^b$ with $a, b \geq 1$, and the weight is $258 + 2^d - 2^{a+1} - 2^{b+1}$. Since $(V_3 \cap V_1) \oplus (V_3 \cap V_2) \subseteq V_3$, we have $a + b \leq d$. The only solutions of $258 + 2^d - 2^{a+1} - 2^{b+1} \in U$ with $1 \leq a \leq b \leq 7$ and $7 \leq d \leq 14$ are $(w, d, a, b) = (322, 7, 4, 4)$ and $(322, 8, 5, 6)$, and both violate $a + b \leq d$.

Case $p \notin F_3$. Then the weight is $254 + 2^d - 2(\alpha + \beta)$, where $\alpha = |F_1 \cap F_3|$ and $\beta = |F_2 \cap F_3|$ are 0 or powers of 2 from 2 to 128. The only solution in U is $w = 334$, $d = 7$, $\{\alpha, \beta\} = \{8, 16\}$. Then $\dim(V_3 \cap V_1) + \dim(V_3 \cap V_2) = 3 + 4 = 7 = \dim V_3$, so $V_3 = (V_3 \cap V_1) \oplus (V_3 \cap V_2)$. Write $F_3 = c + V_3$ and $c = c_1 + c_2$ with $c_i \in V_i$. As F_3 meets F_1 , $c \in V_1 + V_3 = V_1 + (V_3 \cap V_2)$, so $c_2 \in V_3 \cap V_2$; similarly $c_1 \in V_3 \cap V_1$. Hence $c \in V_3$ and $0 = p \in F_3$, a contradiction.

Both statements about solutions are checked by `cert_arith`, and by hand they are quick, as $2^{a+1} + 2^{b+1}$ and $\alpha + \beta$ have at most two binary digits 1. $\square$

The obstruction is dimension counting: in $\text{RM}(6, 12)$ the 6-flats spanned by $e_1, \dots, e_6$, by $e_7, \dots, e_{12}$ and by $e_1, e_2, e_3, e_7, e_8, e_9$ give $162 = 2.5 \cdot 64 + 2$, whereas 322 in $\text{RM}(7, 14)$ would need $a = b = 4$, against $a + b \leq 7$.

5.2 Sums of few monomials

Proposition E. *No sum of at most five monomials of degree at most 7 in $x_1, \dots, x_{14}$ (the constant 1 allowed) has weight in U . The same holds for $f \circ \phi$ for every such sum f and every affine bijection ϕ of $\mathbb{F}_2^{14}$.*

Proof by exhaustive computation. By Lemma 2.1, the weight of $\sum_{i=1}^k x_{S_i}$ depends only on the Venn region sizes n_T , $\emptyset \neq T \subseteq \{1, \dots, k\}$, which satisfy $\sum_T n_T \leq 14$ and $|S_i| = \sum_{T \ni i} n_T \leq 7$. Equal monomials cancel in pairs, so the sums of at most five distinct monomials are exactly the sums $\sum_{i=1}^k x_{S_i}$ with $k \in \{4, 5\}$ and not necessarily distinct S_i . We enumerated all vectors (n_T) for $k \leq 5$: there are 8, 204, 20,708, 8,568,777 and 11,690,447,828 of them for $k = 1, \dots, 5$. Two independent programs (Section 6) agree on these counts, and neither finds a weight in U . The second statement holds because ϕ preserves weights. $\square$

Among these sums, the weights 2 modulo 4 in $[250, 400]$ are 254, 314, 318 and all such numbers from 338 to 398, and the weights below 320 are exactly the elements of K , in agreement with (P4a).

5.3 A reduction

Lemma 5.1. *Let $f \in \text{RM}(7, 14)$ have support S and weight $w \equiv 2 \pmod{4}$. For $a \in \mathbb{F}_2^{14} \setminus \{0\}$ let n_a be the number of pairs $\{x, x + a\}$ contained in S . Then $\sum_{a \neq 0} n_a = \binom{w}{2}$, every n_a is odd, and in coordinates in which a is the last unit vector we have $f = g + x_{14}h$ with $g \in \text{RM}(7, 13)$, $h \in \text{RM}(6, 13)$, $n_a = |\text{supp } g \setminus \text{supp } h|$ and $\text{wt } h = w - 2n_a$.*

Proof. Each 2-subset $\{x, y\}$ of S is counted once, by $a = x + y$. Composing f with an invertible linear map sending e_{14} to a , we may assume $a = e_{14}$. By Lemma 2.2 with $m = 13$ and

$r = 6$, $f = g + x_{14}h$ with $g = f(\cdot, 0) \in \text{RM}(7, 13)$ and $h = f(\cdot, 0) + f(\cdot, 1) \in \text{RM}(6, 13)$; the pairs in direction e_{14} correspond to the points of $\text{supp } g \cap \text{supp}(g + h) = \text{supp } g \setminus \text{supp } h$, and $\text{wt } h = w - 2n_a$. By (P2), $4 \mid \text{wt } h$, so $2n_a \equiv w \equiv 2 \pmod{4}$. $\square$

Proposition F. *Let $w \in U_0$. Then w is a weight of $\text{RM}(7, 14)$ if and only if there are $h \in \text{RM}(6, 13)$ and $g \in \text{RM}(7, 13)$ with*

$$(\text{wt } h, |\text{supp } g \setminus \text{supp } h|) = (w - 2, 1) \quad \text{or} \quad (w - 6, 3).$$

Moreover, a codeword of weight $w = 322, 326, 330, 334$ has such a decomposition, as in Lemma 5.1, in at least 7559, 7235, 6908, 6576 directions.

Proof. If g and h are as stated, then $f = g + x_{14}h \in \text{RM}(7, 14)$ has weight $\text{wt } h + 2|\text{supp } g \setminus \text{supp } h| = w$ by Lemma 2.2. Conversely, let f have weight w . By Lemma 5.1 the n_a are odd and $\sum_{a \neq 0} (n_a - 1) = \binom{w}{2} - 16383$. A direction with $n_a \geq 5$ contributes at least 4 to this sum, so at most $\lfloor (\binom{w}{2} - 16383)/4 \rfloor$ directions have $n_a \geq 5$; for $w = 322, 326, 330, 334$ this bound is 8824, 9148, 9475, 9807. In each of the remaining directions $n_a \in \{1, 3\}$, and Lemma 5.1 gives the decomposition with $\text{wt } h = w - 2n_a$. $\square$

6 Computations

6.1 Certificates

Every computation used in a proof was done by two programs written separately (in `certificates/rm714/` of version 1.4.0 of [5], with data and logs). For Theorems A and B, `verify_witnesses.c` (binary Möbius transform) and `verify_bitset.py` with `verify_all_py.py` (big-integer bit sets) check degrees and weights; both accept the code-words of Table 1, their complements and the 7901 entries of `rm714_all_witnesses.json` (SHA-256 784bca3b...), whose weights are exactly the elements of S_0 . For Lemma 4.1 and Propositions C', D and F, `cert_arith.py` and `cert_arith.c` print the same 38 lines. For Proposition E, `venn_enum.c` (for $k \leq 4$) and `venn_enum_fast.c` (for $k = 5$) recurse over the Venn regions and use Lemma 2.1, while `venn_enum_indep.c` splits the regions monomial by monomial and uses Möbius inversion. Their logs agree on the counts for every k , on all weights up to 400 for $k \leq 4$, and, for $k = 5$, on the weights 2 modulo 4 in $[250, 400]$ and on the number 5676 of distinct weights. No run finds a weight in U . The complete sets of weights, which `venn_enum_indep.c` writes to files, were not compared.

6.2 Searches for the open weights (heuristic)

For fixed $h \in \text{RM}(6, 13)$, the existence of $g \in \text{RM}(7, 13)$ with $|\text{supp } g \setminus \text{supp } h| = n$, for $n = 1$ or 3 as in Proposition F, is linear algebra: let $H = \text{supp } h$ and let $c(x) \in \mathbb{F}_2^{2380}$ be the values at $x \in \mathbb{F}_2^{13}$ of the monomials of degree at most 5, a column of a parity-check matrix of $\text{RM}(7, 13)$, whose dual is $\text{RM}(5, 13)$ [11, Chapter 13]. A vector supported on a set P disjoint from H agrees outside H with a codeword if and only if $\sum_{x \in P} c(x) \in \text{span } c(H)$. The program `closure_search.c` samples structured h of degree at most 6 (sums of two to four pieces, each the indicator of a 7- to 9-flat or the indicator of a 9- to 11-flat times a sum of one to three products of two or three affine forms; or, with the option `GEN_FLATS`, sums of three or four such pieces, mostly indicators of 7- and 8-flats) and decides $n = 1$ and $n = 3$ exactly for each h of a relevant weight. It first maps the $c(x)$ by a fixed random linear map to $\mathbb{F}_2^{512}$, which can create false positives but no false negatives; each positive is re-solved with the full vectors and the codeword is rebuilt and rechecked. No positive was false, and `check_closure_hits.py` rechecks in pure Python every codeword printed in the logs (18 distinct ones for each code). With $m = 11$ the same program runs on $\text{RM}(6, 12)$ as a calibration.

RM(7, 14), $h \in \text{RM}(6, 13)$					RM(6, 12), $h \in \text{RM}(5, 11)$				
w	$n = 1$		$n = 3$		w	$n = 1$		$n = 3$	
	sampld	found	sampld	found		sampld	found	sampld	found
318*	2782 + 549	2782 + 549	4285 + 1010	1360 + 276	158*	3420	3245	4581	1616
322	6574 + 1958	0	2782 + 549	0	162	9581	58	3420	0
326	15 + 89	0	6574 + 1958	0	166	1213	366	9581	58
330	229 + 728	0	15 + 89	0	170	3996	1678	1213	803
334	170 + 587	0	229 + 728	0	174	3052	2820	3996	1679
338*	1159 + 3623	43 + 165	170 + 587	0	178*	32029	886	3052	754
342*	1059 + 2140	239 + 700	1159 + 3623	43 + 165					

Table 2: Searches of Section 6.2: for a target weight w and $n = 1$ (resp. 3), the number of distinct sampled h of weight $w - 2$ (resp. $w - 6$), and the number of them admitting g with $|\text{supp } g \setminus \text{supp } h| = n$, i.e. giving a codeword of weight w . Entries $x + y$: default generator + GEN_FLATS (one h of weight 340 was sampled by both). Starred: known weights, used as controls.

The runs for RM(7, 14) (26 September 2026, 13:59–14:33 UTC) were five of 1200 seconds with the default generator (seeds 11–15), five of 720 seconds with GEN_FLATS (seeds 41–45) and one of 240 seconds (seed 51); the calibration was one run of 120 seconds (seed 21). Consecutive seeds give shifted copies of one random stream, and the runs of each group sampled the same sequence of h . Regenerating the samples from the recorded seeds and iteration counts (`closure_replay.c`) confirms this, and shows that no h of these weights repeats within a run and that the run with seed 51 sampled a subset of the first group. Table 2 counts each h once: 26,956 distinct h of weights 312 to 340, of which 13,681 have the weights 316 to 332 relevant for U_0 .

In RM(6, 12) all four analogues 162, 166, 170, 174 of U_0 were found, and $162 = 2.5 \cdot 64 + 2$ arose only from h of weight exactly $2.5 \cdot 64$, with $n = 1$ (58 of 9581 such h). In RM(7, 14) none of the 8532 sampled h of weight $320 = 2.5 \cdot 128$ admits $n = 1$, and none of the 3331 of weight 316 admits $n = 3$. Earlier, cruder searches found no weight in U_0 either (they did not reach weights near $2^{14} - U_0$): random sparse polynomials (`sparsefind.c`; the weights it found are listed in `sparse_table_s7.txt`), and about $6.7 \cdot 10^9$ random sums of three to six flat indicators and hill-climbing steps (`flatsearch.c`), which met 318 and 338 millions of times. All this is heuristic: the test is exact for each sampled h , but the sampled h are few and special.

7 Formal verification

Theorem A is formalised in `RM714Weights.lean` and `RM714WeightsAudit.lean` (in `lean/Research/` of version 1.4.0 of [5]; Lean 4.33.1, Mathlib [12] commit `0df444a3`). In the namespace `RM714` a polynomial is a list of monomials, a monomial being the set $S \subseteq \{0, \dots, 13\}$ of the indices of its variables (index i stands for x_{i+1}), and `evalPoly L x` is $(\text{L.map fun S} \Rightarrow \text{prod } i \text{ in } S, x \ i). \text{sum}$ in $\mathbb{Z} \text{Mod } 2$. In ASCII notation ($\#$ is the cardinality of a finite set) the other definitions and the main theorem read:

```
def weight (L : List (Finset (Fin 14))) : Nat :=
  #{x : Fin 14 -> ZMod 2 | evalPoly L x = 1}
def IsRM714Weight (w : Nat) : Prop :=
  exists L : List (Finset (Fin 14)), (forall S in L, S.card <= 7) /\ weight L = w
theorem rm714_new_weights :
  forall w in [354, 16030, 4378, 4380, 4382, 8174, 8178, 8182, 8186, 8188,
    8190, 8194, 8196, 8198, 8202, 8206, 8210], IsRM714Weight w
```

So `IsRM714Weight w` says exactly that w is a weight of RM(7, 14) (repeated monomials are allowed; they cancel).

A note on the name. Despite its name, `rm714_new_weights` certifies all 17 weights of Theorem A, of which only 14 are new: 354, 8174, 8178, 8182, 8186, 8188, 8190, 8194, 8196,

8198, 8202, 8206, 8210 and 16030, the elements of $M \setminus U$. The other three, 4378, 4380 and 4382, are asserted by [8, Proposition 4]. The name is historical; the claim of novelty rests on the text of this paper, not on the name of the declaration.

The weights are computed on 14-bit masks: `weightNat` counts the bit patterns at which the exclusive or of the monomials is 1; `weight_map_maskSet` proves that this count is `weight` (using Mathlib’s `finFunctionFinEquiv`); and `isRM714Weight_of` turns masks of degree (`degNat`) at most 7 with count w into `IsRM714Weight w`. The witnesses L354, ..., L8210 (Table 1 and complements) are checked by `isRM714Weight_354`, ..., `isRM714Weight_8210` with `decide +kernel`, that is, by evaluation in the kernel. An audit file prints the definitions and statements and tests the definitions on examples. There is no `sorry`, no `axiom` and no `native_decide`; `#print axioms` for `rm714_new_weights`, `weight_map_maskSet` and `isRM714Weight_of` reports only `propext`, `Classical.choice` and `Quot.sound`. A rebuild with `lake build` in a clean directory (fresh sources, Mathlib build from the local cache) and `leanchecker` on both modules exited with code 0 (26 September 2026, 13:49–13:55 UTC; the log is in version 1.4.0 of the repository). Not formalised are Theorem B (its 7901 codewords would need hours of kernel time in this encoding, and its upper bound rests on [7]), Theorem C, Propositions C’, D, E and F, and the searches.

8 Open questions

Are 322, 326, 330 and 334 weights of $\text{RM}(7, 14)$? This is all that remains of its spectrum, and it decides Carlet’s question for $(c, m) = (7, 14)$. For 322, Proposition F suggests a route: the case $n = 3$ concerns the codewords of weight 316 of $\text{RM}(6, 13)$, described in [7], and for each normal form the existence of g is finite linear algebra; the case $n = 1$ asks whether some $h \in \text{RM}(6, 13)$ of weight exactly 320 has a point $p \notin \text{supp } h$ with $c(p) \in \text{span } c(\text{supp } h)$, as some codewords of weight 160 of $\text{RM}(5, 11)$ do in the calibration.

Tool and computational resource disclosure

This work was carried out on 26 September 2026 (UTC) in a workflow directed by the author, using AI coding agents: Anthropic Claude Code, with the model Claude Opus 5.5. Claude Code agents selected the problem, found the results and their proofs, wrote the programs and the Lean formalisation, carried out the literature searches described in the introduction, and drafted this text; the programs, data, logs and Lean code are in [5]. Throughout, “we” refers to this workflow; the reading and searching described in the paper were done by the agents. The computations ran on one computer (macOS, arm64) on the same day: Proposition E took about 21 CPU minutes with `venn_enum_fast.c` and about 8 minutes on four processes with `venn_enum_indep.c`, the runs behind Table 2 about 2.8 CPU hours, those of `flatsearch.c` 2 hours, and the clean Lean rebuild with `leanchecker` about 5 minutes. The author is not a professional mathematician. The author has read the paper and, using a side-by-side table, compared the statements of the Lean theorems with the results stated here. The results rest on the written proofs, on the codewords of Section 3 with their machine checks, on the computation behind Proposition E and on the known results (P1)–(P4b); the searches of Section 6.2 are heuristic and not used in any proof. AI systems are not authors of this paper; the author takes full responsibility for its content.

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